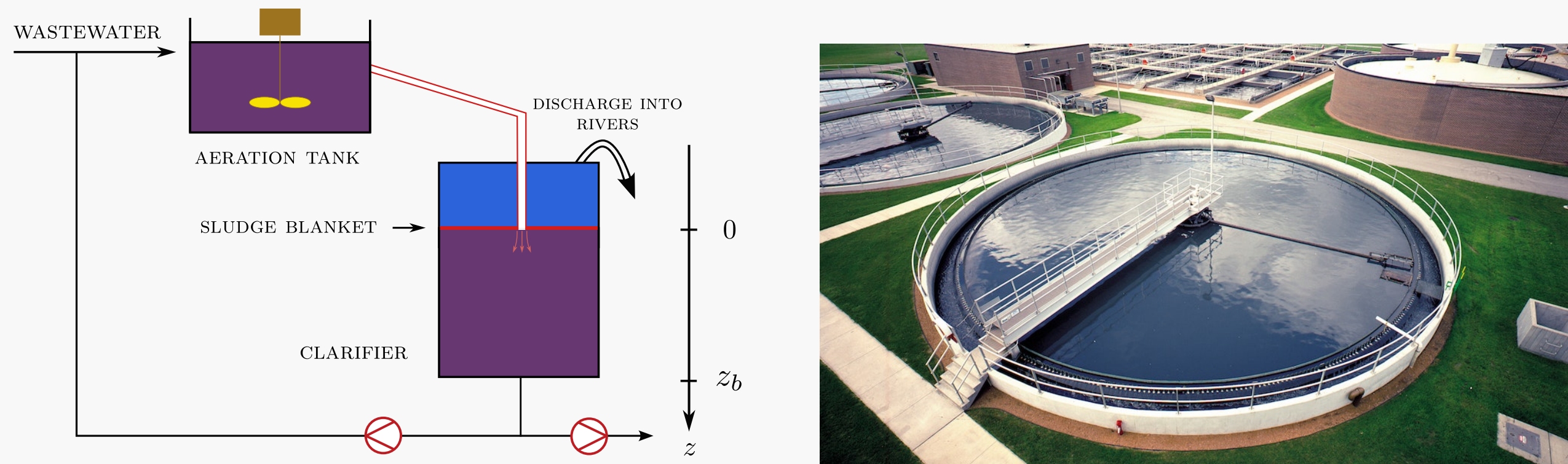


INTRODUCTION AND ENGINEERING CONTEXT



1D-DYNAMIC MODEL OF THE CLARIFIER (TWO-PHASE FLOW)

Let $(t, z) \in [0, +\infty) \times [0, z_b]$. We introduce the conservative variables ε_s (particle volume fraction) and $f_s := \varepsilon_s v_s$ (momentum), where v_s is the particle velocity.

$$\underbrace{\frac{\partial}{\partial t} \begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{F}(\mathbf{Y})} + \underbrace{\frac{\partial}{\partial z} \begin{pmatrix} f_s \\ \frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{S}(\mathbf{Y})} = \underbrace{\begin{pmatrix} 0 \\ \varepsilon_s g \left(1 - \frac{\rho_l}{\rho_s}\right) + \frac{r(\varepsilon_s)(\varepsilon_s \frac{U(Y)}{B} - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} \end{pmatrix}}_{\mathbf{S}(\mathbf{Y})} \quad (1)$$

With an interparticle stress:

$$\sigma_e(\varepsilon_s) = \begin{cases} 0, & \varepsilon_s \leq \varepsilon_c, \\ \sigma_0 \frac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}}, & \varepsilon_s > \varepsilon_c. \end{cases}$$

$$r(\varepsilon_s) = \frac{\rho_l g}{A_k} \varepsilon_s^{\frac{2}{3-n_r}} \quad (\text{liquid/solid drag parameter})$$

$U(Y)$ is the control law.

- ρ_l : water density, g : gravity acceleration
- B : bottom surface area of the Clarifier (m^2)
- $\sigma_0, n_s, n_r, \varepsilon_c, \rho_s$: set of model parameters

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices (pumps).

$$\varepsilon_s(t, 0) = \varepsilon_f, \quad f_s(t, z_b) = \frac{\varepsilon_s(t, z_b)}{B} U(Y)$$

MOTIVATION

Stabilizing the sludge blanket depth requires a stability analysis of the linearized model.

METHODOLOGY AND LINEARIZED MODEL

The linearized system takes the following general form in Riemann coordinates:

$$\begin{cases} \partial_t \phi - \underbrace{\begin{pmatrix} \lambda & c \\ 0 & \lambda \end{pmatrix}}_A \partial_x \phi = 0, \\ \phi(0, x) = \phi^0(x), \quad x \in (0, 1), \end{cases} \quad (2)$$

where $\phi : [0, +\infty) \times [0, 1] \rightarrow \mathbb{R}^2$ is the state, ϕ^0 the given initial condition. $-\lambda < 0$ is the **characteristic** velocity, and $c \neq 0$ is the **coupling** coefficient.

REMARK

The weakly hyperbolic nature of system (2) arises from the fact that the coefficient matrix $-A$ is non-diagonalizable.

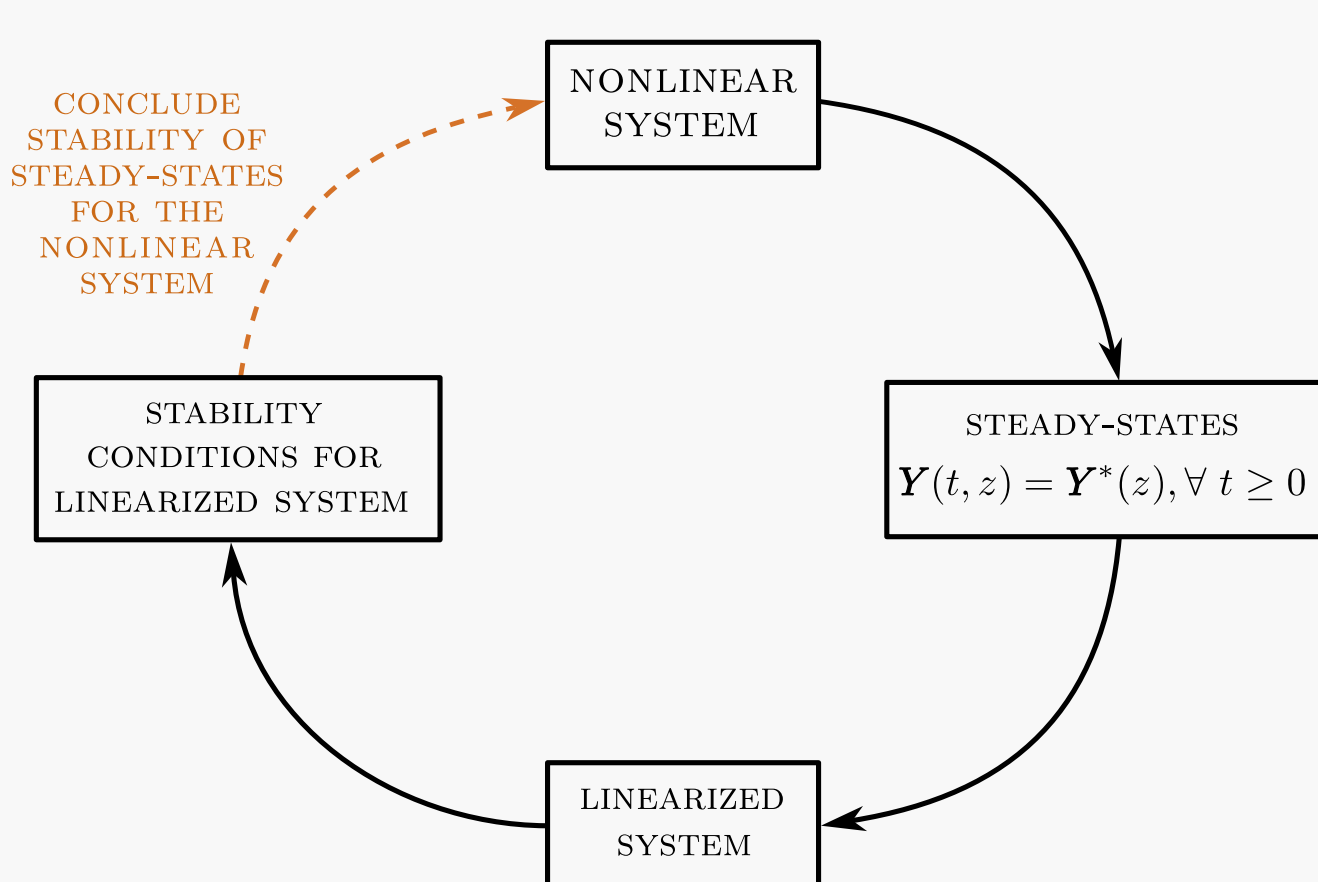


Figure 1. Roadmap on how to stabilize

STABILITY ANALYSIS

We want to identify and characterize **dissipative** boundary conditions which guarantee bounded smooth solutions converging to the steady state $\phi \equiv 0$.

While strictly hyperbolic systems are well-studied for stabilization (see, e.g., Bastin and Coron 2016; Bressan 2000), weakly hyperbolic systems—which appear in various physical contexts (see, e.g., Keyfitz and Kranzer 1980) remain less studied.

WHY LINEAR STATIC BOUNDARY CONDITIONS FAIL?

Assume system (2) is endowed with the following BCs

$$\phi(t, 1) = \begin{pmatrix} 0 & k_1 \\ k_2 & 0 \end{pmatrix} \phi(t, 0), \quad (3)$$

where k_1, k_2 are constant real coefficients.

THEOREM 1 (AMIRAT & AL.)

If the origin of system (2), (3) is globally exponentially stable then $k_2 = 0$.

The proof relies on the representation of system (2), (3) in the Laplace domain ($s \in \mathbb{C}$) which yields the following *characteristic equation* of the poles of the system

$$p_1(z, e^z) \triangleq e^{2z} - \frac{ck_2}{\lambda} z e^z - k_1 k_2 = 0, \quad \text{with } z = \frac{s}{\lambda}.$$

$p_1(z, e^z)$ is a quasipolynomial that does not have a principal term. By Pontryagin's stability results, it has infinitely many roots with *arbitrarily large positive real parts*.

How to relax the constraint $k_2 = 0$?

MIXED (STATIC/DYNAMIC) BOUNDARY CONDITIONS

We propose to replace (3) by the following boundary conditions

$$\phi_1(t, 1) = k_1 \phi_2(t, 0), \quad \frac{d\phi_2}{dt}(t, 1) = k_2 \phi_1(t, 0) - \kappa \phi_2(t, 1), \quad (4)$$

where $\kappa > 0$ is a constant damping coefficient at the boundary $x = 1$.

Remark. System (2), (4) can be written in the form of a NFDE

$$\dot{y}(t) = k_2 \left[k_1 y(t - \frac{2}{\lambda}) + \frac{c}{\lambda^2} \dot{y}(t - \frac{1}{\lambda}) \right] - \kappa y(t), \quad \text{with } y(t) = \phi_2(t, 1).$$

THEOREM 2 (AMIRAT & AL.)

Let $k_2 \neq 0$ and κ satisfy $\kappa > |k_1 k_2|$. We define a critical bound

$$c_{crit, lyap} = \begin{cases} \sqrt{\frac{|k_1|}{|k_2| \kappa}} \lambda^2 \ln\left(\frac{\kappa}{k_1 k_2}\right), & \text{if } \kappa \leq |k_1 k_2| e^2 \\ \frac{2\lambda^2}{e^1 |k_2|}, & \text{if } \kappa > |k_1 k_2| e^2 \end{cases}$$

Then $\forall |c| < c_{crit, lyap}$, the origin of system (2), (4) is globally exponentially stable for the $L^2((0, 1); \mathbb{R}) \times H^1((0, 1); \mathbb{R}) \times \mathbb{R}$ -norm.

The *proof* relies on constructing the following quadratic strict Lyapunov function

$$\mathcal{V} = \alpha_1 \int_0^1 e^{\mu x} \phi_1^2 dx + \alpha_2 \int_0^1 e^{\mu x} \phi_2^2 dx + \alpha_g \int_0^1 e^{\mu x} (\partial_x \phi_2)^2 dx + \frac{1}{2} \beta (\phi_2(t, 1))^2.$$

with strictly positive constant coefficients $\alpha_1, \alpha_2, \alpha_g, \beta, \mu$.

BALANCE LAWS WITH NONUNIFORM COEFFICIENTS

$$\partial_t \phi - \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix} \partial_x \phi = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix} \phi \quad (5)$$

with $c(x) \neq 0, \forall x \in [0, 1], \lambda_{1,2}(x) > 0$ s.t. $\exists x \in [0, 1]$ for which $\lambda_1(x) = \lambda_2(x)$.

Sufficient conditions ensuring exponential stability of system (5), (4)?

- Structural assumption on source: $\exists \mathbf{b} \in C^1([0, 1], \mathbb{R}^*)$, $B_{21}(x) = (\lambda_2(x) - \lambda_1(x)) \mathbf{b}(x)$
- Positivity condition: A linear matrix inequality (LMI) ensuring $V_{Glob} > 0$.
- Dissipation conditions: In-domain and Boundary LMIs ensuring $\dot{V}_{Glob} < -\delta V_{Glob}$.

Here, V_{Glob} is a *non-diagonal quadratic strict Lyapunov function*:

$$\mathcal{V}_{Glob} = \int_0^1 \phi^\top Q(x) \phi dx + \int_0^1 g_1(x) (\partial_x \phi_2)^2 dx + 2 \int_0^1 \mathbf{b}(x) g_1(x) \phi_1 \partial_x \phi_2 dx + \frac{1}{2} \tilde{\beta} \phi_2(t, 1)^2.$$

with $\tilde{\beta} > 0$, $Q \in C^1([0, 1]; \mathbb{S}_2(\mathbb{R}))$ and $g_1 \in C^1([0, 1]; \mathbb{R}^*)$.

Result. System (5)–(4) is globally exponentially stable for the $L^2((0, 1); \mathbb{R}) \times H^1((0, 1); \mathbb{R}) \times \mathbb{R}$ -norm (see Amirat et al. 2026).

CONCLUSIONS/PERSPECTIVES

- We provide a complete framework for stability analysis of linear weakly hyperbolic systems.
- *Ongoing work*: Extension to the nonlinear case (clarifier configuration) in order to stabilize the shock steady-state.

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